

Assignment 4

Data Visualization for EDA

Q1. Matplotlib -- Line Plots, Scatter Plots & Bar Charts

Practice foundational plotting functions in Matplotlib for trend analysis and comparisons.

```
import matplotlib.pyplot as plt
import numpy as np

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 170, 160, 200, 220]
profit = [20, 30, 35, 25, 45, 50]

x = np.array([10, 20, 30, 40, 50, 60, 70, 80])
y = np.array([15, 25, 35, 45, 40, 60, 55, 70])

categories = ['A', 'B', 'C', 'D', 'E']
values = [23, 45, 56, 78, 32]
```

Perform the following tasks:

- Create a line chart showing monthly sales. Add an appropriate title, x-axis label, and y-axis label.
- On the same chart, add a second line showing monthly profit using a different color and line style. Include a legend to distinguish the two lines.
- Create a scatter plot of x versus y. Use red circular markers with black edges and add a grid to the plot.
- Create a vertical bar chart showing values for each category. Use a different color for each bar.
- Create a horizontal bar chart for the same category data.
- Save the horizontal bar chart as an image file named "bar_chart.png" with a resolution of 150 DPI.

Explain in 2-3 sentences: When should you use a line plot versus a scatter plot during EDA?

Q2. Matplotlib -- Histograms, Box Plots & Pie Charts

Visualize data distributions, outliers, and proportions using Matplotlib.

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(42)
data = np.random.normal(loc=50, scale=10, size=1000)
box_data = [np.random.normal(50, 10, 100),
            np.random.normal(60, 15, 100),
            np.random.normal(45, 8, 100)]
labels = ['Group A', 'Group B', 'Group C']

sizes = [30, 25, 20, 15, 10]
explode = (0.1, 0, 0, 0, 0)
```

Perform the following tasks:

- Create a histogram of the data with 30 bins. Use skyblue color with black edges. Add a title "Distribution of Scores".
- Add vertical lines on the histogram to mark the mean and median values in different colors. Include a legend identifying each line.
- Create a box plot comparing the three groups in box_data. Label each group appropriately and display outliers as red diamond markers.
- Create a pie chart showing the proportions in sizes. Label each slice A through E, explode the first slice outward, and display percentage values on each slice.
- Create a stacked bar chart with two data series: [20, 35, 30, 35, 27] and [25, 32, 34, 20, 25]. Use appropriate colors and add a legend.

Explain in 2-3 sentences: What does a box plot reveal that a histogram cannot, and vice versa?

Q3. Matplotlib -- Subplots, Annotations & Customization

Master advanced Matplotlib features for creating publication-ready multi-panel visualizations.

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.cos(x)
y3 = np.tan(x)
y3 = np.clip(y3, -10, 10)

time = np.arange(0, 10, 0.1)
temp = 25 + 10 * np.sin(time) + np.random.normal(0, 2, len(time))
```

Perform the following tasks:

- Create a figure with three subplots arranged vertically (3 rows, 1 column) with a shared x-axis. Set the figure size to 8 inches by 10 inches.
- In the first subplot, plot the sine curve with a blue dashed line. Add an annotation with an arrow pointing to the peak value, labeling it "Peak".
- In the second subplot, plot the cosine curve with a green dotted line. Set the y-axis limits from -1.5 to 1.5.
- In the third subplot, plot the tangent curve with an orange solid line. Add a dashed grid with 70% opacity.
- Apply tight layout to prevent overlapping elements. Add a main title "Trigonometric Functions" at the top of the figure.
- Create a dual-axis plot showing temperature over time. Plot Celsius values on the left y-axis and Fahrenheit values on the right y-axis. Label both axes clearly.

Explain in 2-3 sentences: Why are subplots important when comparing multiple variables in EDA? When would you use a secondary y-axis?

Q4. Seaborn -- Distribution Plots (Univariate Analysis)

Use Seaborn to analyze the distribution of single variables -- the first step in any EDA.

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

np.random.seed(42)
df = pd.DataFrame({
    'Age': np.random.randint(18, 60, 200),
    'Salary': np.random.normal(50000, 15000, 200),
    'Score': np.random.beta(2, 5, 200) * 100,
    'Department': np.random.choice(['IT', 'HR', 'Sales', 'Finance'], 200)
})
```

Perform the following tasks:

- Set the Seaborn visual style to whitegrid and the color palette to muted.
- Create a histogram of the Age column with 20 bins, overlaid with a KDE curve. Use steelblue color.
- Create a KDE plot for the Salary column. Fill the area under the curve with 50% transparency.
- Create a box plot for the Score column. Add a title and ensure all labels are readable.
- Create a violin plot for the Salary column. Display the inner quartile lines inside the violin shape.
- Create an ECDF plot for the Age column. Add a second ECDF line for the Salary column on the same plot (normalize Salary to 0-100 range first, or create separate subplots).

Explain in 2-3 sentences: When would you prefer a violin plot over a box plot for EDA? What unique insight does an ECDF plot provide?

Q5. Seaborn -- Categorical Plots

Analyze relationships between categorical and numerical variables -- essential for understanding group differences.

```
import seaborn as sns
import pandas as pd
import numpy as np

np.random.seed(42)
df = pd.DataFrame({
    'Department': np.random.choice(['IT', 'HR', 'Sales', 'Finance'], 150),
    'Gender': np.random.choice(['Male', 'Female'], 150),
    'Salary': np.random.normal(55000, 12000, 150),
    'Experience': np.random.randint(1, 15, 150)
})
```

Perform the following tasks:

- Create a count plot showing the number of employees in each Department. Use a custom color palette of your choice.
- Create a bar plot showing the average Salary for each Department. Display standard deviation as error bars. Use Gender as a hue to show male and female bars side by side.
- Create a strip plot of Salary by Department with jitter enabled and 60% point opacity. Overlay it on top of a box plot for the same data.
- Create a swarm plot of Salary by Department using the Set2 color palette.
- Create a box plot of Salary by Department, grouped by Gender. Use a pastel color palette.
- Create a violin plot of Experience by Department, split by Gender. Display the median as a white dot inside each half.

Explain in 2-3 sentences: What is the difference between a strip plot and a swarm plot? When should you use each during EDA?

Q6. Seaborn -- Relationship & Regression Plots (Bivariate)

Explore relationships between two continuous variables to find correlations and trends.

```
import seaborn as sns
import pandas as pd
import numpy as np

np.random.seed(42)
df = pd.DataFrame({
    'Study_Hours': np.random.uniform(1, 10, 100),
    'Marks': np.random.uniform(40, 100, 100),
    'Attendance': np.random.uniform(60, 100, 100),
    'Subject': np.random.choice(['Math', 'Science', 'English'], 100)
})
df['Marks'] = df['Study_Hours'] * 5 + df['Attendance'] * 0.3 + np.random.normal(0, 5, 100)
```

Perform the following tasks:

- Create a scatter plot of Study_Hours versus Marks. Color the points by Subject and vary the point size by Attendance.
- Create a line plot showing the relationship between Attendance and Marks. Sort the data by Attendance before plotting and include error shading.
- Create a regression plot of Study_Hours versus Marks with a 95% confidence interval band. Reduce the opacity of the scatter points.
- Create a residual plot for the Study_Hours versus Marks relationship. Add a horizontal reference line at zero in red.
- Create a joint plot of Study_Hours versus Marks with a regression line. Set the plot size to 8 inches.
- Create a pair plot of all numeric columns, colored by Subject. Use KDE plots on the diagonal.

Explain in 2-3 sentences: How does a pair plot help in the initial stages of EDA? What would you look for in the diagonal versus off-diagonal plots?

Q7. Seaborn -- Heatmaps, Clustermaps & Correlation

Visualize correlations and complex matrices to understand multivariate relationships.

```
import seaborn as sns
import pandas as pd
import numpy as np

np.random.seed(42)
df = pd.DataFrame({
    'A': np.random.normal(50, 10, 100),
    'B': np.random.normal(60, 15, 100),
    'C': np.random.normal(40, 8, 100),
    'D': np.random.normal(70, 12, 100),
    'E': np.random.normal(55, 20, 100)
})
df['B'] = df['A'] * 0.8 + np.random.normal(0, 5, 100)
df['C'] = df['A'] * (-0.6) + np.random.normal(0, 5, 100)
df['D'] = df['B'] * 0.5 + np.random.normal(0, 5, 100)
```

Perform the following tasks:

- Compute and print the correlation matrix of all numeric columns in the dataframe.
- Create a heatmap of the correlation matrix. Display the correlation values inside each cell formatted to 2 decimal places. Add grid lines between cells.
- Use a diverging colormap centered at zero with values ranging from -1 to 1.
- Create a clustermap of the correlation matrix. Set the figure size to 8 by 8 inches and use an appropriate diverging colormap.
- Create a heatmap showing only the lower triangle of the correlation matrix. Mask the upper triangle so it appears blank.
- Create a FacetGrid showing the distribution of column A, split into separate subplots based on quartile ranges of column D.

Explain in 2-3 sentences: Why is visualizing correlations via a heatmap more effective than reading a raw correlation matrix? When would you use a clustermap instead?

Q8. Plotly -- Interactive Basic Charts

Build interactive visualizations using Plotly Express for dynamic data exploration.

```
import plotly.express as px
import pandas as pd
import numpy as np

np.random.seed(42)
df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'] * 3,
    'Sales': np.random.randint(100, 300, 18),
    'Profit': np.random.randint(20, 80, 18),
    'Region': ['North', 'South', 'East'] * 6,
    'Customers': np.random.randint(50, 200, 18)
})
```

Perform the following tasks:

- Create an interactive line chart showing Sales over Month, with a separate line for each Region. Include markers at each data point.
- Create an interactive bar chart showing total Sales by Region. Display the exact sales value on top of each bar.
- Create an interactive scatter plot of Sales versus Profit. Color points by Region, set bubble size by Customers, and show Month information when hovering over points.
- Create an interactive donut chart showing the proportion of total Sales by Region. Set the hole size to 40% of the radius.
- Create an animated scatter plot of Sales versus Profit that transitions through each Month. Fix the x-axis range from 0 to 350 and y-axis range from 0 to 100 for smooth animation.
- Create an interactive area chart showing cumulative Sales by Month for each Region.

Explain in 2-3 sentences: What advantage does Plotly offer over static Matplotlib/Seaborn plots during EDA? Give a specific example.

Q9. Plotly -- Graph Objects, 3D Plots & Controls

Use Plotly Graph Objects for custom interactive visualizations with full control over layout and behavior.

```
import plotly.graph_objects as go
import pandas as pd
import numpy as np

np.random.seed(42)
df = pd.DataFrame({
    'Product': ['A', 'B', 'C', 'D', 'E'] * 4,
    'Quarter': ['Q1', 'Q2', 'Q3', 'Q4'] * 5,
    'Revenue': np.random.randint(50, 200, 20),
    'Cost': np.random.randint(30, 150, 20)
})
df['Profit'] = df['Revenue'] - df['Cost']
```

Perform the following tasks:

- Create a grouped bar chart showing Revenue and Cost side by side for each Product. Use appropriate colors and add a title.
- Create a stacked bar chart of the same Revenue and Cost data. Ensure the total height of each bar represents the sum.
- Create a bubble chart where the x-axis shows Revenue, the y-axis shows Profit, bubble size represents Cost, and bubble color represents Product.
- Add interactive dropdown buttons that allow the user to toggle the bar chart between showing Revenue, Cost, or Profit values.
- Create a 3D scatter plot with Revenue on the x-axis, Cost on the y-axis, and Profit on the z-axis. Color points by Product and label all three axes.
- Create an interactive table displaying the first 10 rows of the dataframe with formatted column headers.

Explain in 2-3 sentences: When would you choose Plotly Graph Objects over Plotly Express? What is the trade-off?

Q10. Plotly -- Interactive EDA Dashboards & Advanced Charts

Build advanced interactive visualizations specifically designed for EDA -- hierarchical charts, faceted views, and multi-plot dashboards.

```
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
import pandas as pd
import numpy as np

np.random.seed(42)
n = 300
df = pd.DataFrame({
    'City': np.random.choice(['New York', 'LA', 'Chicago', 'Houston', 'Phoenix'], n),
    'Category': np.random.choice(['Electronics', 'Clothing', 'Food', 'Home'], n),
    'Sales': np.random.lognormal(4, 0.5, n),
    'Profit': np.random.lognormal(3, 0.4, n),
    'Rating': np.random.choice([1, 2, 3, 4, 5], n, p=[0.05, 0.1, 0.2, 0.35, 0.3]),
    'Quarter': np.random.choice(['Q1', 'Q2', 'Q3', 'Q4'], n)
})
```

Perform the following tasks:

- Create an interactive sunburst chart showing the hierarchy from City to Category, with Sales as the value. Color the segments by Profit using a color scale.
- Create an interactive treemap with the same City-to-Category hierarchy. Display the average Rating when hovering over each segment.
- Create a faceted scatter plot of Sales versus Profit with a separate subplot for each Quarter. Color points by Category.
- Create a 2x2 multi-plot dashboard containing: (1) a histogram of Sales, (2) a box plot of Profit by Category, (3) a scatter plot of Sales versus Profit, and (4) a bar chart of average Sales by City. Add subplot titles.
- Create an animated bar chart showing Sales by Category that transitions through each Quarter. Fix the y-axis range to accommodate the maximum value across all quarters.
- Create an interactive violin plot of Sales by Category, grouped by City. Display a box plot inside each violin and show all individual data points.

Explain in 2-3 sentences: How do interactive dashboards (multi-plot layouts) improve EDA compared to viewing one chart at a time? When is a sunburst or treemap more useful than a bar chart?